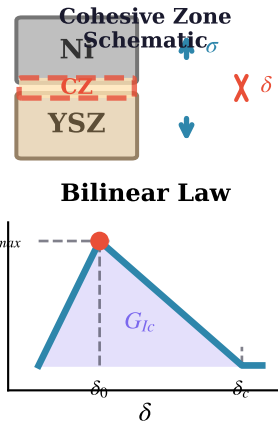
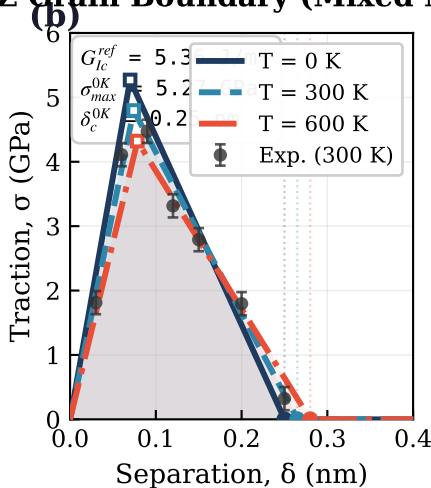
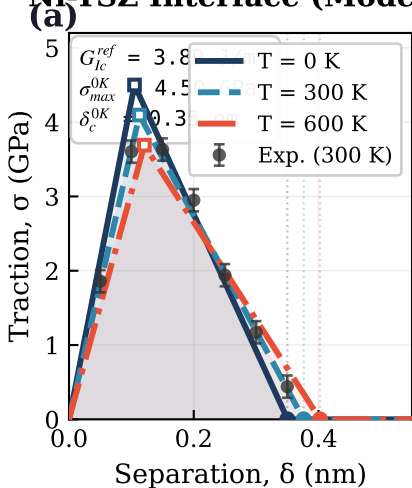
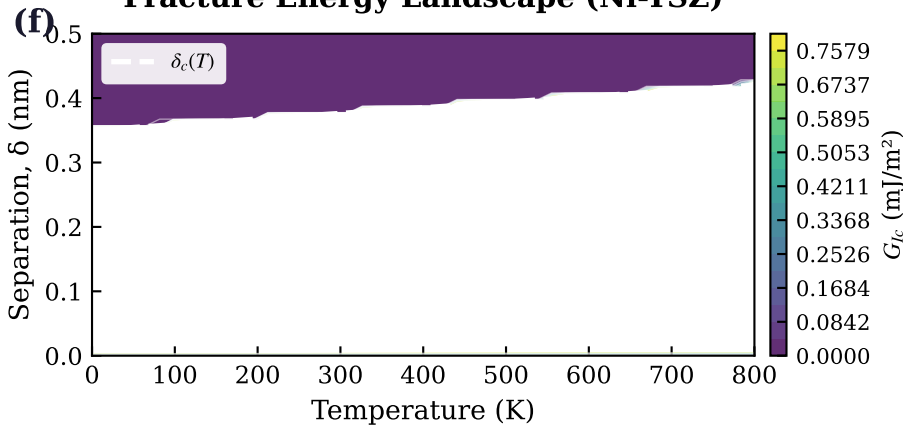
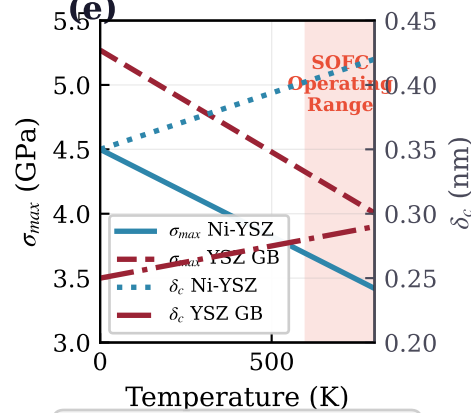
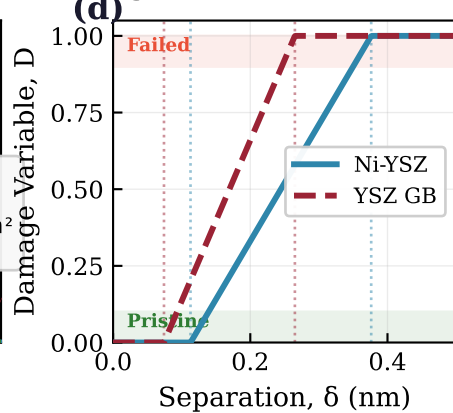
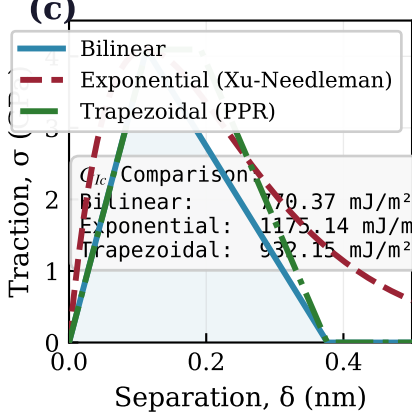


Cohesive Zone Traction-Separation Laws for SOFC Fracture Modeling

Ni-YSZ Interface (Mode I) YSZ Grain Boundary (Mixed Mode)



Cohesive Law Formulations (T = 300 K) Damage Evolution (T = 300 K) Temperature Dependence



KEY FINDINGS (0K → 600K)

Ni-YSZ Interface:

σ_{max} reduction: 18.0%
 δ_c increase: 15.0%

YSZ Grain Boundary:

σ_{max} reduction: 18.0%
 δ_c increase: 12.0%

Temperature dependence enables early decohesion at SOFC operating temps